

DFSMS Basics: Introduction to ABARS

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Today's Agenda

- Basic Terminology
- ABARS Introduction
- Defining the aggregate
- ABACKUP Highlights
- ARECOVER Highlights
- ABARS Utilities

OS/390 Terminology

- **SMF** = System Management Facility
 - ▶ Trace records for activity in the OS/390 system
- **ISMF** = Interactive Storage Management Facility (panels for DFSMS)
- **GDPS** = Geographically-Dispersed Parallel Sysplex
- **PPRC** = Peer-to-Peer Remote Copy
- **XRC** = Extended Remote Copy

DFSMS Terminology

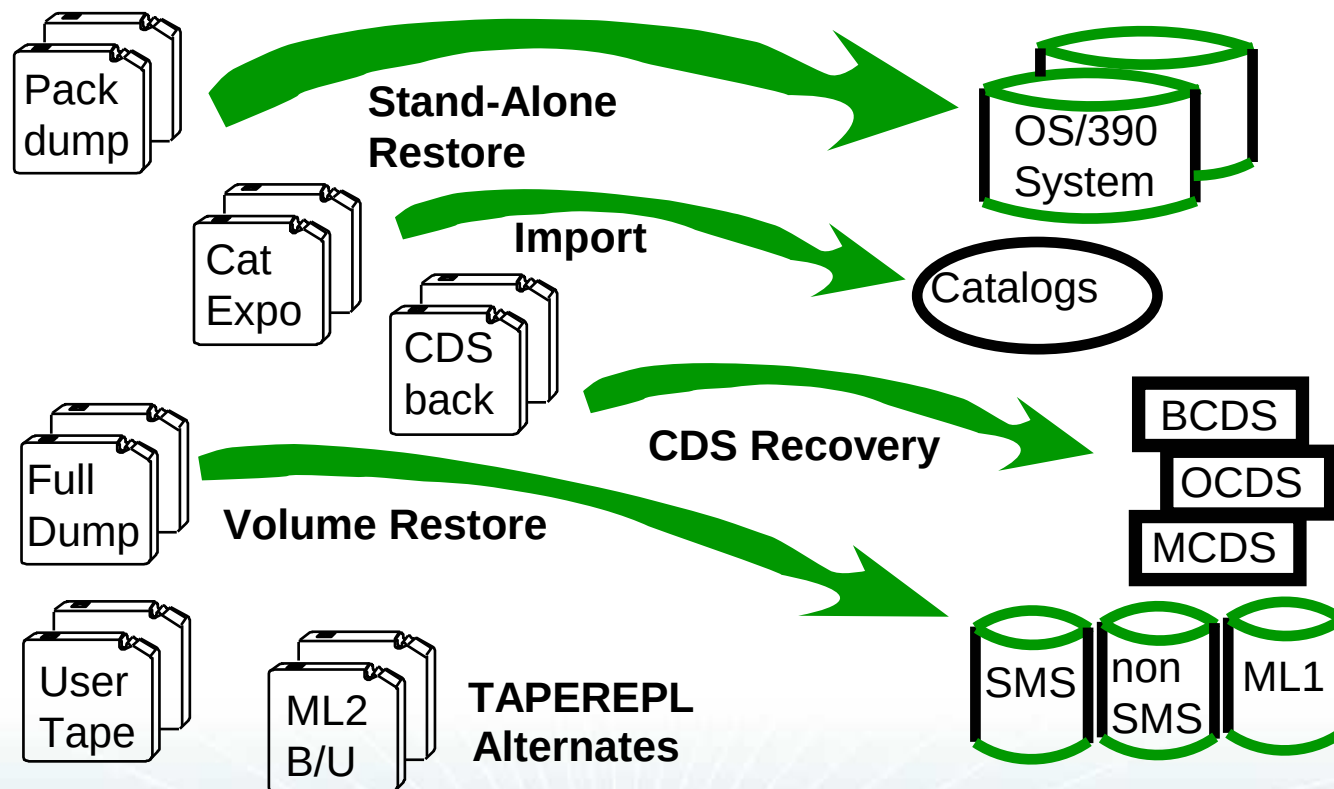
- **DFSMS** = Data Facility Storage Management Subsystem
- **RMM** = Removable Media Manager
- **SMS** = System-Managed Storage
- **ACS** = Automatic Class Selection (routines)
- **ICF** = Integrated Catalog Facility
- **GDG** = Generation Data Group
- **GDS** = Generation Data Set

DFSMSHsm Terminology

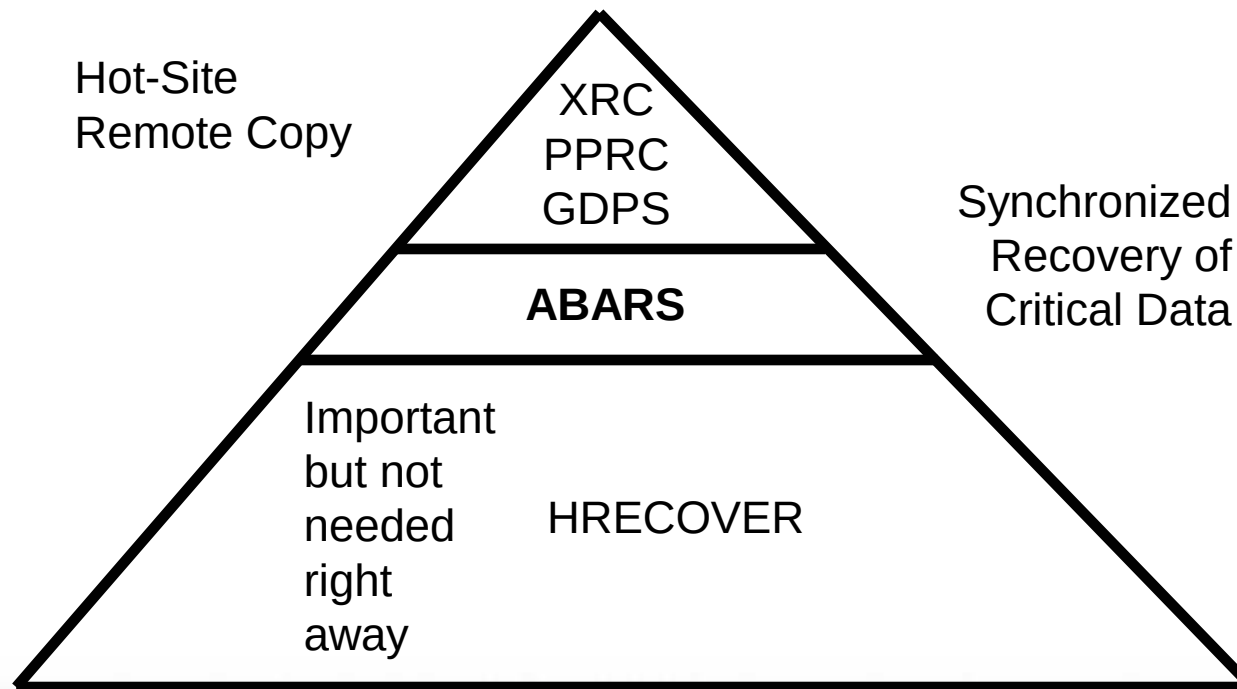


- **HSM** = Hierarchical Storage Manager
- **CDS** = Control Data Set (MCDS, BCDS, OCDS)
- **ML1** = Migration Level 1, DASD volumes
- **ML2** = Migration Level 2, usually tape
- **SDSP** = Small Data Set Packing data set used to consolidate small migrated data sets on ML1 volume

Disaster Recovery - One Approach



Where does ABARS fit in ?



ABARS Strengths



- Device Independence
- Reduces storage requirements
 - ▶ **Recovery site cost savings**
- Handles both DASD and TAPE data sets
- Process migrated data without recall
- Simple data set selection criteria
- Command or panel driven function



ABARS Strengths

- Extended data set conflict resolution options
- Prioritize recovery of applications
- Concurrent copy support
- Allows local testing to whatever devices available
- Rename during recovery
- Can be used to define ICF User Catalogs

ABARS Strengths

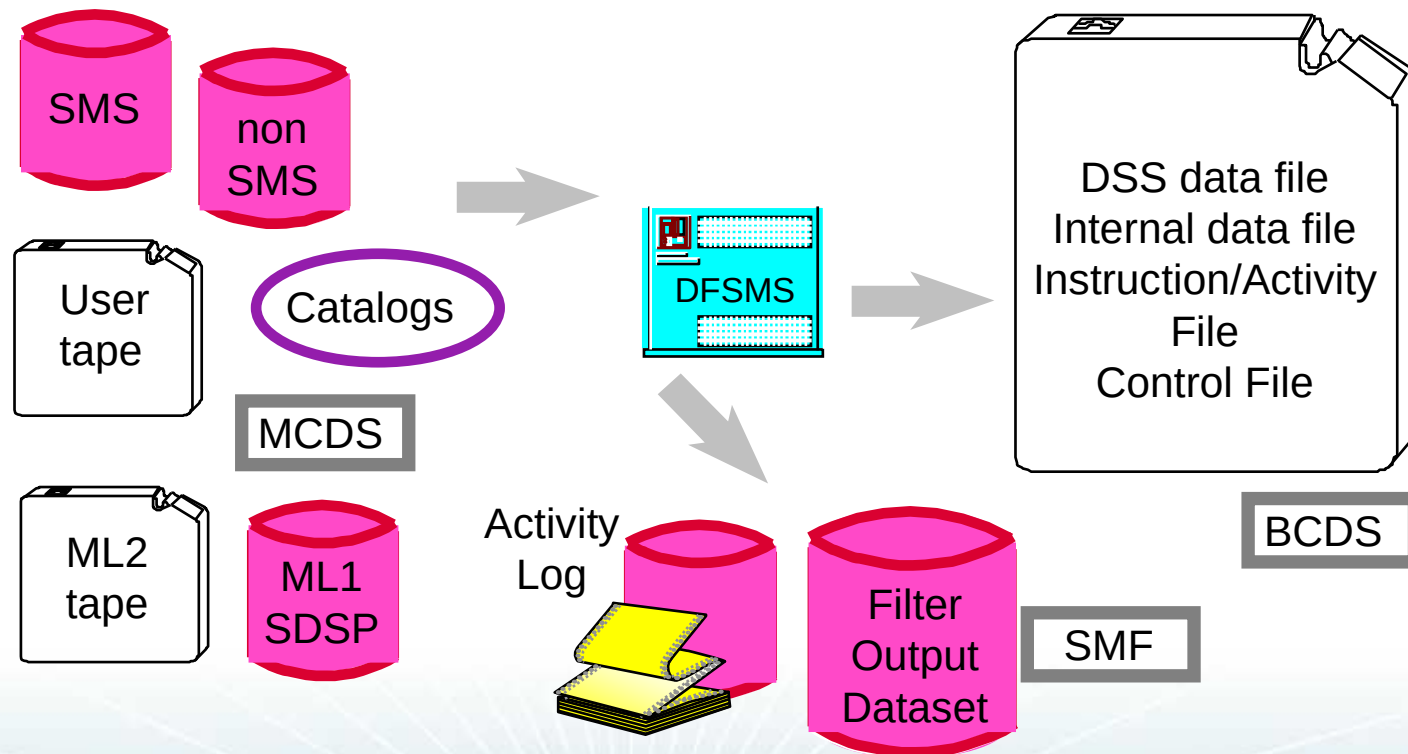
■ Data Integrity

- ▶ All data sets backed up at one point in time
- ▶ Catalog Information Taken
- ▶ CDS information taken for migrated data
- ▶ GDG base information taken for GDG data sets
 - ▶ *With or without backing up generations*
- ▶ *Eliminates out of synch conditions with your data*

ABARS Introduction

- **ABARS** - Aggregate backup and recovery support
 - ▶ Function of DFSMSHsm
- **Aggregate** = a group of data sets and their associated control information
- Backup and recovery at aggregate level
- **Logical** dump and restore
- Focus on critical data

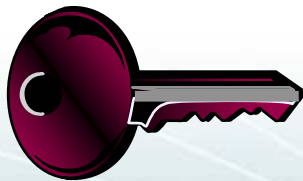
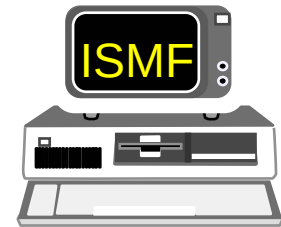
The ABACKUP process



Defining the Aggregate



- Aggregate defined by Storage Administrator
- Aggregate group is an SMS construct
- Aggregate attributes are:
 - ▶ Aggregate group name (**1-8 characters**)
 - ▶ Outputdataset prefix (**1-35 characters**)
 - ▶ Number of copies (**1-15**)
 - ▶ ***Selection data set*** name(s)



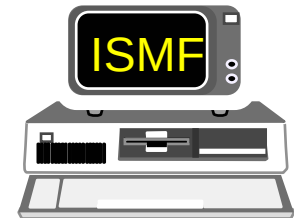
This is the key to ABARS!

Defining the Aggregate



- Aggregate attributes (continued):

- ▶ Instruction data set name
- ▶ Management class name (**1- 8 characters**)
- ▶ Description (**1-120 characters**)
- ▶ Accounting Information (**1-32 characters**)

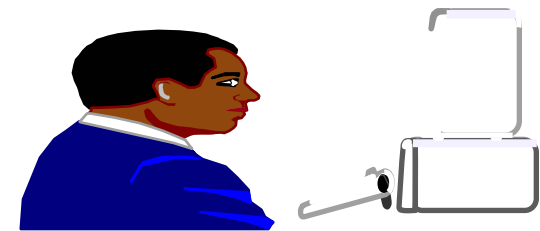


- SMS Configuration **must** be activated after modifying the definition

- Selection and instruction data sets can be modified at any time **without reactivation**

Define Selection Dataset(s)

- Where data set lists are maintained:
- 3 sets of keywords:
 - ▶ INCLUDE / EXCLUDE
 - *Data sets to be copied*
 - ▶ ALLOCATE / ALLOCATEEXCLUDE
 - *Data sets to be allocated empty (DASD)*
 - ▶ ACCOMPANY / ACCOMPANYEXCLUDE
 - *Data sets to be cataloged only (TAPE)*

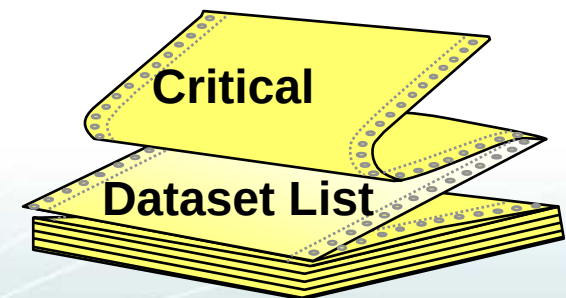


Application Owner

Define Selection Dataset(s)

■ SYNTAX

- ▶ Fully or Partially qualified
- ▶ Relative GDG names
- ▶ Partially qualified makes **aggregate dynamic**
- ▶ GDG bases retrieved for all GDSs
- ▶ ICF User Catalogs and GDG base names can be in ALLOCATE list
 - ***All associated Aliases obtained***



Define Management Class

- Must be defined if management class name specified in aggregate group definition.
- If management class **not** specified ABARS will use either default management class or DFSMSHsm defaults.
- Use ISMF panels or NaviQuest jobs to define Management Classes and Aggregate Groups

Define Management Class



ISMP SHARE.ORG

- The aggregate attributes are:
 - ▶ Number of versions ([1-9999](#) | [NOLIMIT](#))
 - ▶ Retain ONLY version
 - (days | weeks | months | years | NOLIMIT)
 - ▶ Retain EXTRA versions
 - ▶ COPY SERIALIZATION
 - [Continue or Fail](#)
 - ▶ ABACKUP COPY TECHNIQUE
 - [Standard](#) | [Preferred](#) | [Required](#)

ABACKUP Highlights

- **VERIFY** option - to test syntax of selection data set and validate setup.
- Backs up migrated data in compressed format.
- Obtains GDG base definition for GDGs.
- Can backup ICF Usercatalog definition and associated aliases
- **PROCESSONLY** keyword to select specific hierarchy levels (L0, ML1, ML2, UserTape)

ABACKUP Highlights

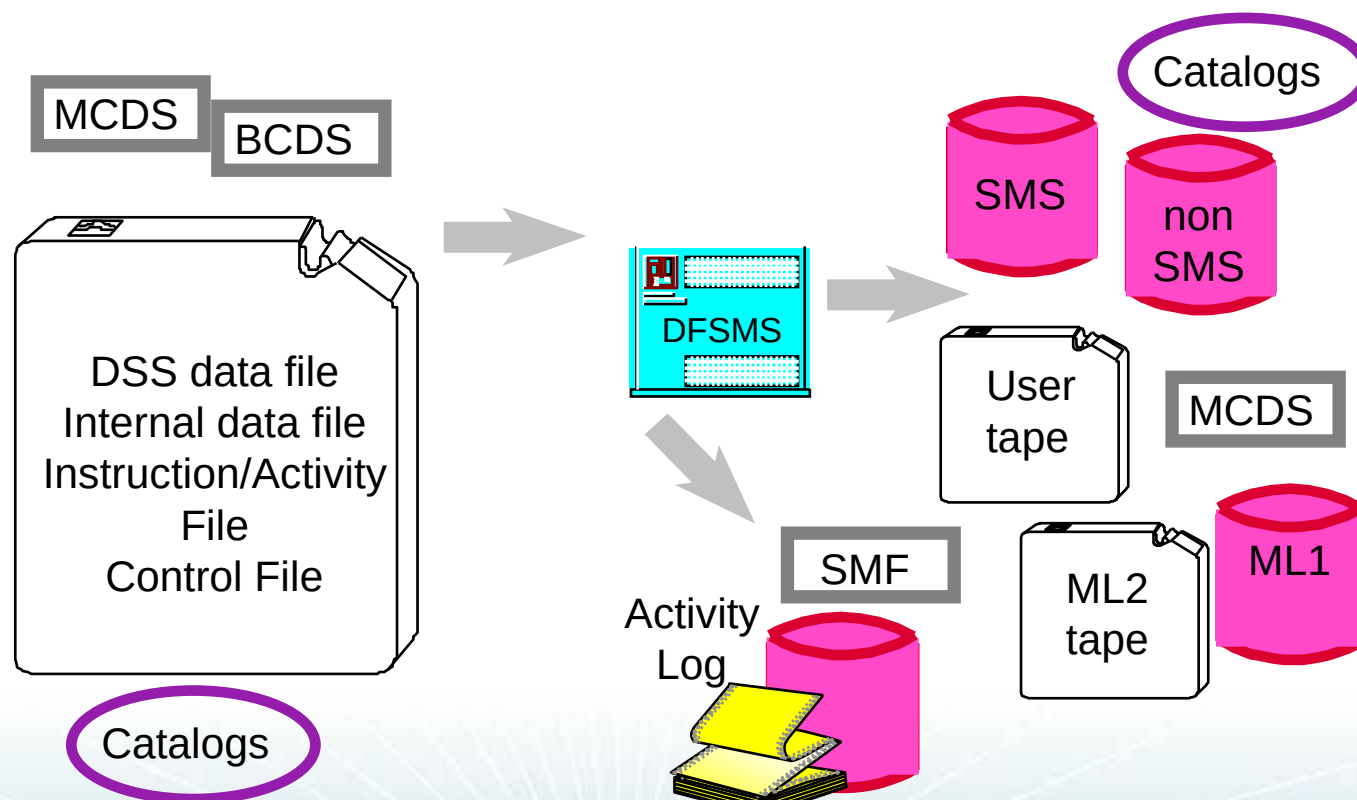
- Create record in the DFSMSHsm BCDS to track ABACKUP inventory
- Automatic "roll-off" processing
- Can redirect output to DASD via ACS routines.
- Reports amount of storage backed up from each level of hierarchy.
- **Stacking** minimizes number of output cartridges required per aggregate

ABACKUP Highlights



- Supports backup to 3592 WORM tape

The ARECOVER process



ARECOVER Highlights

- **PREPARE** option - sets up aggregate recovery environment.
- **VERIFY** option - to test validity of aggregate recovery and checks for data set conflicts.
- **RESTARTABLE** - remembers what data sets have been successfully recovered.

ARECOVER Highlights



- Can detect "*orphan catalog entries*" when performing conflict resolution
 - ▶ Data set in catalog but volume not online
 - ▶ Data set in catalog, volume online, but data set not on VTOC
 - ▶ Data set migrated but no associated CDS record
- Can define ICF User Catalogs and Aliases
- Can define GDG base definition, if needed

Data Set Conflict Resolution

■ **REPLACE**

- Delete Target
- Recover Source

■ **RENAMETARGET**

■ **RENAMESOURCE**

- Recover Source with different name

■ **BYPASS**

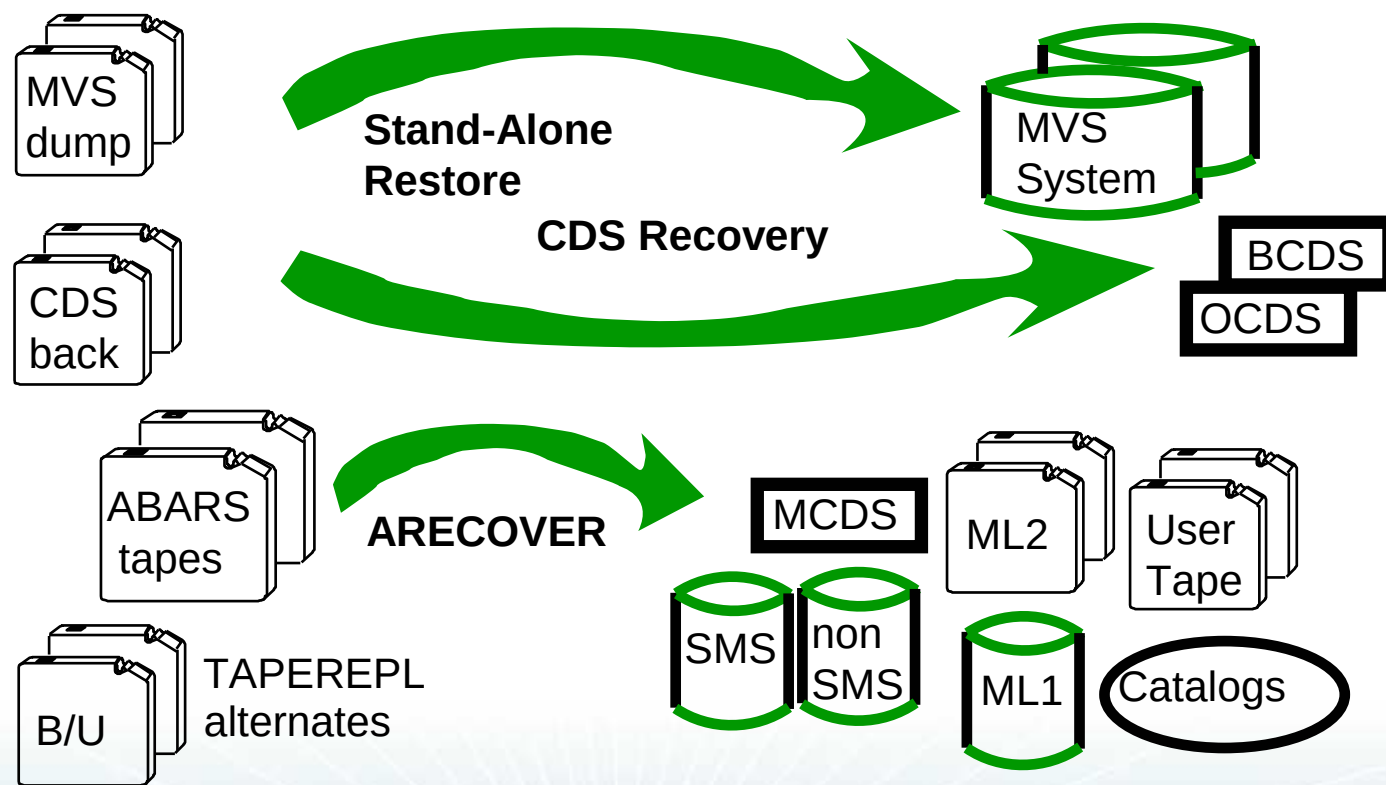


ARECOVER Highlights



- Can recover migrated data to ML1, ML2, or SOURCELEVEL.
- ACS routines invoked during data set allocation.
 - ▶ Recovery site environment determines if data is SMS managed or not

Disaster Recovery - ABARS Approach



ABARS Utilities



■ ABARS Utilities

- ▶ Shipped in SYS1.SAMPLIB(ARCTOOLS)
- ▶ Assists in managing ABARS
- ▶ Provides ABARS data set inventory
- ▶ Allows for individual data set restore
- ▶ Builds ARECOVER command syntax

Summary



- Reviewed ABARS terms and concepts
- Highlighted features and functions of ABACKUP and ARECOVER
- Reviewed other tools that might be useful with ABARS